Conservation of Mass and Momentum for the Flow of Interdendritic Liquid during Solidification

S. GANESAN and D.R. POIRIER

In this paper, mass and momentum conservation equations are derived for the flow of interdendritic liquid during solidification using the volume-averaging approach. In this approach, the mushy zone is conceived to be two interpenetrating phases; each phase is described with the usual field quantities, which are continuous in that phase but discontinuous over the entire space. On the microscopic scale, the usual conservation equations along with the appropriate interfacial boundary conditions describe the state of the system. However, the solution to these equations in the microscopic scale is not practical because of the complex interfacial geometry in the mushy zone. Instead, the scale at which the system is described is altered by averaging the microscopic equations over some representative elementary volume within the mushy zone, resulting in macroscopic equations that can be used to solve practical problems. For a fraction of liquid equal to unity, the equations reduce to the usual conservation equations for a single-phase liquid. It is also found that the resistance offered by the solid to the flow of interdendritic liquid in the mushy zone is best described by two coefficients, namely, the inverse of permeability and a second-order resistance coefficient. For the flow in columnar dendritic structures, the second-order coefficient along with the permeability should be evaluated experimentally. For the flow in equiaxial dendritic structures (*i.e.*, isotropic media), the inverse of permeability alone is sufficient to quantify the resistance offered by the solid.

I. INTRODUCTION

It is well known that the motion of the interdendritic liquid during solidification is often the primary cause of macrosegregation.[1] Recognizing this fact, Flemings *et al.*[2,3,4] analyzed macrosegregation caused by the flow of solute-rich interdendritic liquid because of the volume changes during solidification. Mehrabian *et al.*[5] were the first to consider the effect of buoyancy as well as volume changes during solidification on the flow of interdendritic liquid. In that work, as well as in some that followed,[6,7,8] all of the investigators viewed the mushy zone as a porous medium and determined the velocities of interdendritic liquid using Darcy's law. As will be apparent later in this paper, Darcy's law represents flow of interdendritic liquid in some limited situations, and, hence, the results based solely on Darcy's law are also restricted.

Szekely and Jassal[9] developed a model for dendritic solidification in which they also viewed the mushy zone as a porous medium; they represented the flow of interdendritic liquid with a modified Navier-Stokes equation in which they replaced the viscous term by the resistance term in Darcy's equation. Very recently, Bennon and Incropera[10] developed a continuum model to represent the transport processes in the mushy zone during solidification. Their model is based on a mixture theory in which the mushy zone is viewed as an overlapping continuum. That is, any location in the mushy zone can be occupied by solid and liquid simultaneously. On this basic premise, they derived the conservation equations for the mushy zone, which "are an intuitive extension of classical single-phase relations," according to Hassanizadeh and Gray.[11] With their approach, one has to include intuitively the interactions between phases. For example, they represented the forces resulting from liquid-solid interactions in the mushy zone by the Darcy's term.

In this paper, we use the volume-averaging technique to derive the macroscopic conservation equations for mass and momentum in the mushy zone during solidification. The mushy zone comprises two interpenetrating phases; each phase is described "microscopically" with the usual field quantities which are continuous in that phase but discontinuous over the entire space. On the "microscopic" scale, the usual conservation equations, along with the appropriate interfacial boundary conditions, describe the state of the system. The solution of these equations in the microscopic scale is not practical because of the highly complex interfacial geometries in the mushy zone; thus, the scale at which the system is described is altered by averaging the microscopic equations over some representative elementary volume. This results in a set of "macroscopic" equations that can be used to solve practical problems. As examples, they have been applied to flow in porous media,[12,13,14] fluidized beds,[15,16] solid-fluid mixtures,[17] flow of immiscible fluids,[18] and drying.[19]

In our review of the literature for an acceptable momentum conservation equation for the flow of interdendritic liquid in the mushy zone, we found seven different representations.[9,10,20–25] This indicated to us the need for a systematic approach of deriving a rigorous representation of the conservation of momentum in the mushy zone. Very recently, Beckermann[20] and Beckermann and Viskanta[25] used the volume-averaging technique to derive the macroscopic equations for the mushy zone, but the explicit details and underlying assumptions are not given in their derivations, in the context of mushy zone solidification, although Beckermann

S. GANESAN, Graduate Student, and D.R. POIRIER, Professor, are with the Department of Materials Science and Engineering, The University of Arizona, Tucson, AZ 85721.

Manuscript submitted February 16, 1989.

provides reference to appropriate volume-averaging literature. Hence, it is also our objective to show the details of the derivation of the mass and momentum conservation equations for the mushy zone during solidification.

II. AVERAGING PROCEDURE

Consider a mushy zone during solidification. In the macroscopic field, every unit element in the mushy zone has a centroid within an averaging volume, dV; an averaging area, dA, which bounds the averaging volume, is also defined. Based on the continuum hypothesis, the average quantities should be continuous functions of space and time and should be independent of the size of the averaging volume. In Figure 1, the mass average density of the interdendritic liquid is plotted as a function of the characteristic size of the averaging volume at a particular time. In this context, the mass average density of the liquid is defined as the mass of the liquid in the averaging volume divided by the entire volume of the averaging volume. From Figure 1, we can see that when dV is very small, the average mass density of the interdendritic liquid varies discontinuously, depending on whether the centroid of dV lies in the liquid or solid. This region is represented by the dotted lines. As the size of dV increases, the average density fluctuates continuously, because portions of liquid or solid phases may become included in dV. Eventually, these fluctuations diminish as the size of dV increases, and within a certain characteristic size of dV, the density or any other average quantities become independent of dV. This region is represented by the horizontal line in Figure 1. However, if the characteristic size of dV is too large, the global variations in these properties of the system (mushy zone) appear, as indicated schematically in Figure 1 for a characteristic length greater than d. Based on this argument, the characteristic length of dV or dA should satisfy the following restriction:

$$l \ll d \ll L \qquad [1]$$

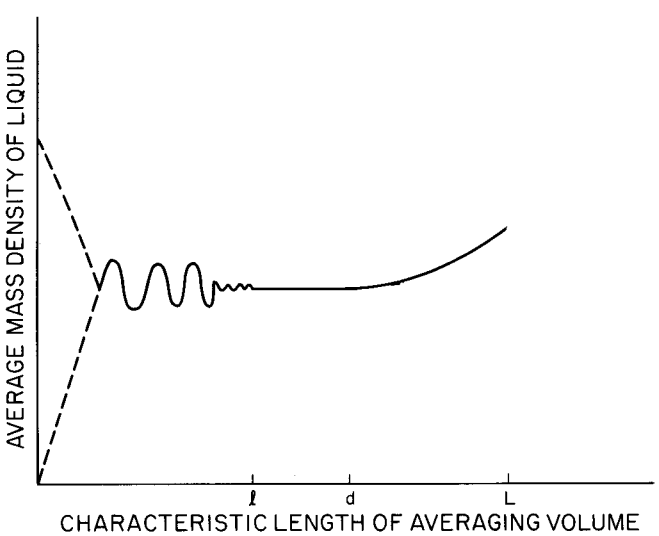

Fig. 1—Computed density as a function of the characteristic length of the averaging volume.

where l is the microscopic characteristic length; d is the characteristic length of the averaging volume (*e.g.*, several dendrite arm spacings); and L is the characteristic length of the mushy zone (*e.g.*, width of the mushy zone). From Eq. [1], we can say that the averaging volume dV represents a physical point in the macroscopic field (mushy zone). Sometimes dV is called the representative elementary volume (REV); similarly, the averaging area, dA, is called the representative elementary area (REA).[11]

The position of the centroid of an averaging volume with respect to a fixed frame of reference is given by $\mathbf{x}$. If within dV, the position vector of a point with respect to the fixed frame of reference is $\mathbf{r}$ and with respect to the centroid of dV it is $\boldsymbol{\xi}$, then (Figure 2)

$$\mathbf{r} = \mathbf{x} + \boldsymbol{\xi} \qquad [2]$$

The volume fraction of the liquid phase is defined as g_l; it is the fraction of dV occupied by the liquid phase. Thus,

$$g_l(\mathbf{x}, t) = \frac{dV_l}{dV}(\mathbf{x}, t) = (1/dV) \int_{dV} \gamma \, dv \qquad [3]$$

where t is time, $dV_l(\mathbf{x}, t)$ is the part of dV occupied by interdendritic liquid, dv is the microscopic unit element (represents $\boldsymbol{\xi}$ coordinate), and $\gamma(\mathbf{r}, t)$ is the liquid volume distribution function, whose value is 1 for a point in the liquid and 0 for a point in the solid. Similarly, the areal fraction of the liquid phase is defined as

$$S_l(\mathbf{x}, t) = \frac{dA_l}{dA}(\mathbf{x}, t) = (1/dA) \int_{dA} \gamma(\mathbf{r}, t) \, da \qquad [4]$$

where dA_l is the part of dA occupied by the liquid and da is the microscopic elemental area. Following the arguments of Whitaker,[26]

$$S_l = g_l \qquad [5]$$

Any microscopic quantity, ψ, has a macroscopic value (*i.e.*, its volume average). The macroscopic value of ψ

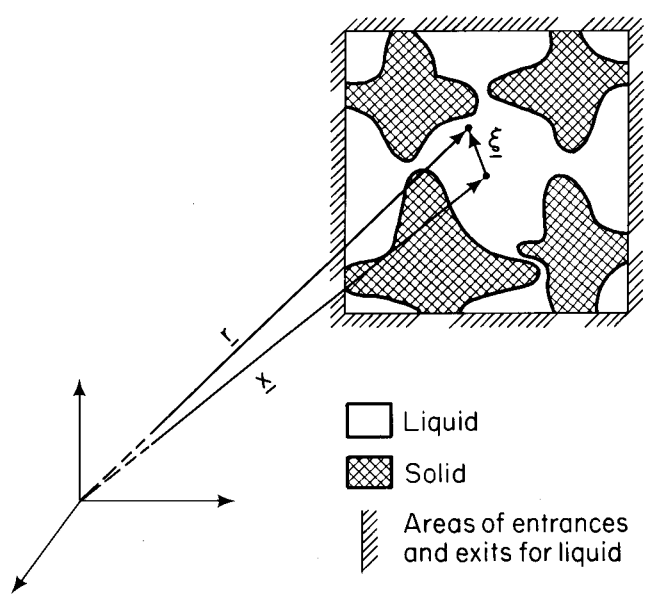

Fig. 2—The averaging volume in the mushy zone.

associated with the interdendritic liquid is Ψ_l and is defined as

$$\Psi_l(\mathbf{x}, t) = (1/dV) \int_{dV} \psi_l(\mathbf{r}, t)\, \gamma(\mathbf{r}, t)\, dv \qquad [6]$$

The intrinsic volume average of ψ_l is defined as

$$\Psi_l^l(\mathbf{x}, t) = \Psi_l / g_l \qquad [7]$$

Sometimes, it is convenient and appropriate to work with the mass average of any quantity. The mass average is defined as

$$\bar{\Psi}_l(\mathbf{x}, t) = (1/\rho_l^l g_l dV) \int_{dV} \rho_l(\mathbf{r}, t) \psi_l(\mathbf{r}, t) \gamma\, dv \qquad [8]$$

where ρ_l is the mass density of the liquid and ρ_l^l is the intrinsic volume average mass density of the liquid and is given by

$$\rho_l^l(\mathbf{x}, t) = (1/g_l dV) \int_{dV} \rho_l(\mathbf{r}, t) \gamma\, dv \qquad [9]$$

In dendritic freezing processes, the local temperature and concentration of solute are uniform[1] and coupled by the liquidus curve (or surface) of the equilibrium phase diagram. Thus, within the averaging volume dV, $\rho_l(\mathbf{r}, t)$ is uniform so that $\rho_l^l = \rho_l$. When this condition is met, the intrinsic volume average of ψ_l and the mass average of ψ_l are equal; thus,

$$\bar{\Psi}_l = \Psi_l^l = \Psi_l / g_l \qquad [10]$$

III. AVERAGING THEOREMS AND IDENTITIES

The following averaging theorems and identities are given by several independent workers,[11,26-29] and these form the basis for the volume-averaging techniques. They are repeated here because they are applied in the following sections in deriving the macroscopic conservation equations for the mushy zone.

A. *Theorem 1*

This relates the average of the spatial derivative to the spatial derivative of the average

$$(1/dV) \int_{dV_l} (\nabla \psi_l)\, dv = \nabla \left[(1/dV) \int_{dV_l} \psi_l\, dv \right] + (1/dV) \int_{a^*} \psi_l \mathbf{n}_{ls}\, da^* \qquad [11]$$

where a^* is the interfacial area between the liquid and solid within dV and $\mathbf{n}_{ls}$ is the outward unit normal of the microscopic element of area da^* pointing from the liquid toward the solid. Equation [11] can also be written using γ as follows:

$$(1/dV) \int_{dV} (\nabla \psi_l) \gamma\, dv = \nabla \left[(1/dV) \int_{dV} \psi_l \gamma\, dv \right] + (1/dV) \int_{a^*} \psi_l \mathbf{n}_{ls}\, da^* \qquad [12]$$

Similar expressions can be written for any microscopic quantity in solid phase (ψ_s).

B. *Theorem 2*

This relates the average of a time derivative to the time derivative of the average

$$(1/dV) \int_{dV_l} (\partial \psi_l / \partial t)\, dv = \frac{\partial}{\partial t} \left[(1/dV) \int_{dV_l} \psi_l\, dv \right] - (1/dV) \int_{a^*} \psi_l \mathbf{w} \cdot \mathbf{n}_{ls}\, da^* \qquad [13]$$

where $\mathbf{w}$ is the velocity of the microscopic liquid-solid interface. Using the liquid volume distribution function, Eq. [13] becomes

$$(1/dV) \int_{dV} (\partial \psi_l / \partial t) \gamma\, dv = \frac{\partial}{\partial t} \left[(1/dV) \int_{dV} \psi_l \gamma\, dv \right] - (1/dV) \int_{a^*} \psi_l \mathbf{w} \cdot \mathbf{n}_{ls}\, da^* \qquad [14]$$

There are also four identities[11] involving the deviation of a microscopic quantity at point $\mathbf{r}$ from its mass average value at $\mathbf{x}$; recall that $\mathbf{x}$ gives the position of the centroid of dV. The deviation of a microscopic quantity from its mass-averaged quantity is denoted by

$$\tilde{\psi}_l(\mathbf{r}, t) = \psi_l(\mathbf{r}, t) - \bar{\Psi}_l(\mathbf{x}, t) \qquad [15]$$

Because $\bar{\Psi}_l(\mathbf{x}, t)$ is constant within dV, the first identity is

$$\bar{\tilde{\psi}}_l = 0 \qquad [16]$$

where $\bar{\tilde{\psi}}_l$ is the mass average of $\tilde{\psi}_l$.

The second identity is

$$\overline{\tilde{\psi}_l \bar{\Phi}_l} = 0 \qquad [17]$$

where $\bar{\Phi}_l$ is the mass average of some other microscopic quantity ϕ_l. The long overbar encompassing the product of the two variables indicates that the mass average is taken after the product.

The third identity is

$$\overline{\psi_l \phi_l} = \bar{\Psi}_l \bar{\Phi}_l + \overline{\tilde{\psi}_l \tilde{\phi}_l} \qquad [18]$$

where $\overline{\tilde{\psi}_l \tilde{\phi}_l}$ is commonly called a "dispersion quantity."

Finally, the fourth identity is

$$(1/dV) \int_{dV} \rho_l \tilde{\phi}_l \bar{\Psi}_l \gamma\, dv = 0 \qquad [19]$$

IV. CONSERVATION OF MASS FOR THE MUSHY ZONE

Consider the microscopic mass conservation equation for the interdendritic liquid:

$$\frac{\partial \rho_l}{\partial t} + \nabla \cdot \rho_l \mathbf{v}_l = 0 \qquad [20]$$

where $\mathbf{v}_l$ is the microscopic velocity of the liquid. Multiplying Eq. [20] by γ and averaging over dV yields

$$(1/dV)\int_{dV}(\partial\rho_l/\partial t)\gamma\,dv + (1/dV)\int_{dV}(\nabla\cdot\rho_l\mathbf{v}_l)\gamma\,dv = 0 \qquad [21]$$

Using Theorem 2 for the first term in Eq. [21] and Theorem 1 for the second term, and realizing that the microscopic density is uniform in dV, Eq. [21] becomes

$$\frac{\partial}{\partial t}(\rho_l g_l) + \nabla\cdot(\rho_l g_l\bar{\mathbf{V}}_l) + (1/dV)\int_{a^*}\rho_l(\mathbf{v}_l - \mathbf{w})\cdot\mathbf{n}_{ls}\,da^* = 0 \qquad [22]$$

where $\bar{\mathbf{V}}_l$ is the mass average velocity of the interdendritic liquid (*i.e.*, the macroscopic velocity of the interdendritic liquid) as defined by Eq. [8]. Clearly, $(\mathbf{v}_l - \mathbf{w})\cdot\mathbf{n}_{ls}$ represents the normal component of the relative microscopic velocity of the interdendritic liquid because of the volume change accompanying solidification. A similar equation can be obtained for the solid phase as

$$\frac{\partial}{\partial t}(\rho_s g_s) + \nabla\cdot[\rho_s g_s\bar{\mathbf{V}}_s] + (1/dV)\int_{a^*}\rho_s(\mathbf{v}_s - \mathbf{w})\cdot\mathbf{n}_{sl}\,da^* = 0 \qquad [23]$$

where $\mathbf{v}_s$ is the microscopic velocity of solid, $\bar{\mathbf{V}}_s$ is the mass average velocity of solid, ρ_s is the microscopic density of solid in dV, which is assumed equal to the intrinsic volume average density (ρ_s^s), g_s is volume fraction solid, and $\mathbf{n}_{sl}$ is the unit outward normal to da^* from the solid toward the liquid. Note that the mass gained by the solid because of solidification is equal to the mass lost by liquid (or *vice versa*). Thus, the area integrals in Eqs. [22] and [23] are equal in magnitude and opposite in sign, and the integrals cancel when Eqs. [22] and [23] are added. The result is

$$\frac{\partial}{\partial t}(\rho_s g_s + \rho_l g_l) + \nabla\cdot\rho_l g_l\bar{\mathbf{V}}_l + \nabla\cdot\rho_s g_s\bar{\mathbf{V}}_s = 0 \qquad [24]$$

For a stationary solid, $\mathbf{v}_s = \bar{\mathbf{V}}_s = 0$, and Eq. [24] becomes

$$\frac{\partial}{\partial t}(\rho_s g_s + \rho_l g_l) + \nabla\cdot\rho_l g_l\bar{\mathbf{V}}_l = 0 \qquad [25]$$

In terms of the average interdendritic liquid velocity, $\mathbf{V}_l^l$ (intrinsic volume average) or the superficial velocity, $\mathbf{V}_l$ (volume average), Eq. [25] can be written as

$$\frac{\partial}{\partial t}(\rho_s g_s + \rho_l g_l) + \nabla\cdot\rho_l g_l\mathbf{V}_l^l = 0 \qquad [25a]$$

or

$$\frac{\partial}{\partial t}(\rho_s g_s + \rho_l g_l) + \nabla\cdot\rho_l\mathbf{V}_l = 0 \qquad [25b]$$

respectively. Because ρ_l is uniform within dV, $\bar{\mathbf{V}}_l = \mathbf{V}_l^l$ in deriving Eq. [25a] from Eq. [25]. It is reassuring to note that Eq. [25a] agrees with the conservation of mass in the mushy zone, as given first by Flemings and Nereo in 1967.[2]

Only if the densities of the solid and liquid are equal in dV, uniform throughout the mushy zone, and constant can Eq. [25b] be written as

$$\nabla\cdot\mathbf{V}_l = 0 \qquad [26]$$

using $g_s + g_l = 1$. Beckermann[20] also derived Eqs. [25b] and [26]. However, he did not explicitly state that the densities of the solid and liquid are equal in dV and that the density of the liquid is globally uniform throughout the mushy zone in deriving Eq. [26]. Beckermann also assumed that the velocities attributable to density changes upon solidification are negligible. As can be seen, however, it is not necessary to neglect the velocities attributed to solidification shrinkage in order to obtain Eq. [26].

V. CONSERVATION OF MOMENTUM FOR THE MUSHY ZONE

If the solid is stationary, the microscopic momentum conservation equation for the liquid can be written as[30]

$$\frac{\partial}{\partial t}(\rho_l\mathbf{v}_l) + \nabla\cdot\rho_l\mathbf{v}_l\mathbf{v}_l = -\nabla p_l + \rho_l\mathbf{g} + \nabla\cdot\sigma_l^{(2)} \qquad [27]$$

where $\mathbf{g}$ is gravitational acceleration, p_l is mechanical pressure, and $\sigma_l^{(2)}$ is the deviatoric part of the general material stress tensor of the liquid element. Multiplying Eq. [27] by γ and averaging over dV, we get

$$(1/dV)\int_{dV}\frac{\partial}{\partial t}(\rho_l\mathbf{v}_l)\gamma\,dv + (1/dV)\int_{dV}(\nabla\cdot\rho_l\mathbf{v}_l\mathbf{v}_l)\gamma\,dv = -(1/dV)\int_{dV}(\nabla p_l)\gamma\,dv + (1/dV)\int_{dV}\rho_l\mathbf{g}\gamma\,dv + (1/dV)\int_{dV}(\nabla\cdot\sigma_l^{(2)})\gamma\,dv \qquad [28]$$

A. *The Transient and Inertial Terms*

Consider the left-hand side (LHS) of Eq. [28] and apply Theorem 2 for the first term and Theorem 1 for the second term. Also, with Eqs. [8] and [15] through [19] for the second term, the LHS of Eq. [28] becomes

$$(1/dV)\int_{dV}\frac{\partial}{\partial t}(\rho_l\mathbf{v}_l)\gamma\,dv + (1/dV)\int_{dV}(\nabla\cdot\rho_l\mathbf{v}_l\mathbf{v}_l)\gamma\,dv = \frac{\partial}{\partial t}(\rho_l g_l\bar{\mathbf{V}}_l) + \nabla\cdot(\rho_l g_l\bar{\mathbf{V}}_l\bar{\mathbf{V}}_l) + \nabla\cdot\left[(1/dV)\int_{dV}\rho_l\bar{\mathbf{v}}_l\bar{\mathbf{v}}_l\gamma\,dv\right] + (1/dV)\int_{a^*}\rho_l\mathbf{v}_l(\mathbf{v}_l - \mathbf{w})\cdot\mathbf{n}_{ls}\,da^* \qquad [29]$$

Equation [29] is obtained by using the fact that ρ_l and

$\bar{\mathbf{V}}_l$ are uniform within dV. The area integral in Eq. [29] represents the exchange of momentum because of the volume change during solidification. When considering convection in a mushy zone where the interdendritic liquid is the major phase (*e.g.*, $0.7 \leq g_l \leq 1$), the liquid motion induced by the thermal and solutal buoyancy forces is much greater than the motion induced by volume changes.[31] On the other hand, for lesser relative amounts of interdendritic liquid, the form and drag forces associated with the motion of the interdendritic liquid, relative to the position of the solid, are certainly much greater than the normal interfacial forces represented as the area integral in Eq. [29]. In a mushy zone, therefore, we safely ignore this term in the momentum balance. In fact, this term can be ignored in solid-liquid mixtures, where the volume change of the transformation or reaction is small; it is significant only in solid-gas or liquid-gas systems.

The third term on the right-hand side (RHS) of Eq. [29] is the "average momentum dispersion vector," which arises because of averaging the nonlinear convective term in the microscopic momentum equation. Remember that $\tilde{\mathbf{v}}_l$ represents the deviation of the microscopic velocity from the mass-averaged velocity in dV. For the flow of liquid in the mushy zone, there are velocity gradients in the microscopic scale, because the liquid motion has to satisfy the no-slip condition on the liquid-solid interface and because the interdendritic spaces are tortuous. This gives rise to the deviation of the microscopic velocity from the mass-averaged velocity. The contribution of this term to the overall momentum balance can be understood by considering the two limiting cases for g_l in a mushy zone. As $g_l \rightarrow 1$, $\bar{\mathbf{V}}_l \rightarrow \mathbf{v}_l$ and $\tilde{\mathbf{v}}_l \rightarrow \mathbf{0}$; thus, this term is insignificant when compared to the convective term (*i.e.*, the second term on the RHS of Eq. [29]). For small values of g_l, the liquid-solid interaction forces certainly dominate the overall momentum balance, so this term can be ignored. However, for some critical range of g_l values, perhaps this term may contribute significantly to the overall momentum balance in the mushy zone. Nevertheless, with this reservation, this term is neglected in the subsequent analysis, and the LHS of Eq. [28] becomes

$$(1/dV)\int_{dV}\frac{\partial}{\partial t}(\rho_l\mathbf{v}_l)\gamma\,dv + (1/dV)\int_{dV}(\nabla\cdot\rho_l\mathbf{v}_l\mathbf{v}_l)\gamma\,dv = \frac{\partial}{\partial t}(\rho_l g_l\bar{\mathbf{V}}_l) + \nabla\cdot(\rho_l g_l\bar{\mathbf{V}}_l\bar{\mathbf{V}}_l) \qquad [30]$$

B. *The Pressure and Body Force Terms*

Now return to the RHS of Eq. [28]. By making use of Theorem 1 and Eqs. [6] and [7] for the first term and Eq. [3] for the second term, the pressure and the body force terms become

$$-(1/dV)\int_{dV}(\nabla p_l)\gamma\,dv + (1/dV)\int_{dV}\rho_l\mathbf{g}\gamma\,dv = -\nabla(g_l P_l^l) + \rho_l g_l\mathbf{g} - (1/dV)\int_{a^*}p_l\mathbf{n}_{ls}\,da^* \qquad [31]$$

Equation [31] is obtained by again recognizing that ρ_l is constant in dV. The term P_l^l represents the average pressure (*i.e.*, the intrinsic volume average) acting on the liquid in dV.

The area integral in Eq. [31] is the normal force acting on the solid and can be decomposed into two parts. One part is attributable to the relative motion between the liquid and the solid which is the "form drag" ($\mathbf{F}_D$).[32] The other part must satisfy the mechanical equilibrium between the solid and the liquid within dV. Therefore, $-\mathbf{F}_s/dV$ cancels $P_l^l\nabla g_l$, after the pressure term is expanded, so that only $g_l\nabla P_l^l$ and $\mathbf{F}_D/dV$ remain.

C. *The Viscous Force Terms*

Consider the last term on the RHS of Eq. [28]. Using Theorem 1, we get

$$(1/dV)\int_{dV}(\nabla\cdot\sigma_l^{(2)})\gamma\,dv = \nabla\cdot\left[(1/dV)\int_{dV}\sigma_l^{(2)}\gamma\,dv\right] + (1/dV)\int_{a^*}\sigma_l^{(2)}\cdot\mathbf{n}_{ls}\,da^* \qquad [32]$$

The first term on the RHS of Eq. [32] represents the liquid-liquid interaction forces. Liquid can interact with liquid in the other part of the mushy zone only through the areas of entrances and exits, dA_l (Figure 2). The second term pertains to the liquid-solid interaction forces.

Assuming that the liquid is Newtonian, the constitutive relation between stresses and strain rates (velocity gradients) can be written as[33]

$$\sigma_l^{(2)} = \mu[\nabla\mathbf{v}_l + (\nabla\mathbf{v}_l)^T] - \left(\frac{2}{3}\right)\mu(\nabla\cdot\mathbf{v}_l)I^{(2)} \qquad [33]$$

where μ is the viscosity of liquid; $I^{(2)}$ is the second-order unit tensor; and the superscript T represents the transpose. Substituting Eq. [33] into the first term of the RHS of Eq. [32] and recognizing that viscosity is constant within dV (but viscosity varies globally) give

$$\nabla\cdot\left[(1/dV)\int_{dV}\sigma_l^{(2)}\gamma\,dv\right] = \nabla\cdot\left[(\mu/dV)\int_{dV}(\nabla\mathbf{v}_l)\gamma\,dv\right] + \nabla\cdot\left[(\mu/dV)\int_{dV}(\nabla\mathbf{v}_l)^T\gamma\,dv\right] - \left(\frac{2}{3}\right)\nabla\cdot\left[(\mu/dV)\int_{dV}(\nabla\cdot\mathbf{v}_l)I^{(2)}\gamma\,dv\right] \qquad [34]$$

For the first term on the RHS of Eq. [34], Theorem 1 is used:

$$\nabla\cdot\left[(\mu/dV)\int_{dV}(\nabla\mathbf{v}_l)\gamma\,dv\right] = \nabla\cdot\left[\mu\nabla\left\{(1/dV)\int_{dV}\mathbf{v}_l\gamma\,dv\right\} + (\mu/dV)\int_{a^*}\mathbf{v}_l\mathbf{n}_{ls}\,da^*\right] \qquad [35]$$

The area integral vanishes because of the no-slip condition on the interface, the solid is stationary, and the volume change is negligibly small. By applying Eq. [6] for the integral in the first term, Eq. [35] becomes

$$\nabla \cdot \left[(\mu/dV) \int_{dV} (\nabla \mathbf{v}_l)\gamma \, dv \right] = \nabla \cdot [\mu(\nabla \mathbf{V}_l)] \quad [36]$$

In terms of average interdendritic liquid velocity or mass-averaged velocity, Eq. [36] can be written as

$$\nabla \cdot \left[(\mu/dV) \int_{dV} (\nabla \mathbf{v}_l)\gamma \, dv \right] = \nabla \cdot [\mu \nabla (g_l \mathbf{V}_l^l)] \quad [36a]$$

or

$$\nabla \cdot \left[(\mu/dV) \int_{dV} (\nabla \mathbf{v}_l)\gamma \, dv \right] = \nabla \cdot [\mu \nabla (g_l \bar{\mathbf{V}}_l)] \quad [36b]$$

Using the same procedure, the other terms of Eq. [34] can be written in terms of the mass-averaged velocity as follows:

$$\begin{aligned} \nabla \cdot \left[(1/dV) \int_{dV} \sigma_l^{(2)} \gamma \, dv \right] &= \nabla \cdot [\mu \nabla (g_l \bar{\mathbf{V}}_l) + \mu \{\nabla (g_l \bar{\mathbf{V}}_l)\}^T \\ &\quad - (2\mu/3) \{\nabla \cdot (g_l \bar{\mathbf{V}}_l)\} I^{(2)}] \end{aligned} \quad [37]$$

Now we combine $\mathbf{F}_D$ remaining from the third term on the RHS of Eq. [31] with the second term on the RHS of Eq. [32] to get the total average force exerted by the liquid on the solid in dV, due entirely to the relative motion between liquid and solid in dV. It is

$$\mathbf{F} = (1/dV) \left[-\mathbf{F}_D + \int_{a^*} \sigma_l^{(2)} \cdot \mathbf{n}_{ls} \, da^* \right] \quad [38]$$

The first term in Eq. [38] is the "form drag," and the integral is the "viscous drag" from Eq. [32].

Following the arguments of Whitaker[29] and Gray and O'Neill,[34] for Newtonian fluids, it is assumed that $\mathbf{F}$ depends on the relative mass-averaged velocities between the liquid and solid, and if macroscopically there is no motion, then the net force will be zero. Mathematically,

$$\mathbf{F} = -\mu \mathbf{\Gamma} (\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s) \quad [39]$$

where $\mathbf{\Gamma}$ is commonly called the "tortuosity function."[29,34] Note that there is a negative sign on the RHS of Eq. [39] because of the fact that $\mathbf{F}$ resists the liquid motion in dV. We can expand $\mathbf{\Gamma}(\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s)$ using the Taylor series from $\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s = \mathbf{0}$; this gives

$$\begin{aligned} \mathbf{\Gamma}(\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s) &= \mathbf{R}^{(2)} \cdot (\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s) \\ &\quad + C^{(3)} : (\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s)(\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s) \\ &\quad + \text{higher order terms} \end{aligned} \quad [40]$$

where $R^{(2)}$ and $C^{(3)}$ are second- and third-order tensor resistance coefficients which are functions of the morphology of the solid in the mushy zone (*i.e.*, dendrite arm spacing, volume fraction solid, *etc.*).

The first term on the RHS of Eq. [40] represents the first-order resistance, and the second term is a second-order resistance. Note that $C^{(3)}$ is symmetric in the last two indices because $(\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s)(\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s)$ is symmetric. When the higher order terms are neglected, Eq. [40] can be substituted into Eq. [38] to give

$$\mathbf{F} = -\mu [R^{(2)} \cdot (\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s) + C^{(3)} : (\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s)(\bar{\mathbf{V}}_l - \bar{\mathbf{V}}_s)] \quad [41]$$

For a stationary solid, $\mathbf{v}_s = \mathbf{0}$, and, hence, $\bar{\mathbf{V}}_s = \mathbf{0}$, so that Eq. [41] becomes

$$\mathbf{F} = -\mu [R^{(2)} \cdot \bar{\mathbf{V}}_l + C^{(3)} : (\bar{\mathbf{V}}_l)(\bar{\mathbf{V}}_l)] \quad [42]$$

Putting all terms together, the macroscopic momentum conservation equation for the mushy zone is

$$\begin{aligned} \frac{\partial}{\partial t}(\rho_l g_l \bar{\mathbf{V}}_l) + \nabla \cdot (\rho_l g_l \bar{\mathbf{V}}_l \bar{\mathbf{V}}_l) &= -g_l \nabla P_l^l + \rho_l g_l \mathbf{g} + \nabla \cdot [\mu \nabla (g_l \bar{\mathbf{V}}_l) \\ &\quad + \mu \{\nabla (g_l \bar{\mathbf{V}}_l)\}^T \\ &\quad - (2\mu/3) \{\nabla \cdot (g_l \bar{\mathbf{V}}_l)\} I^{(2)}] \\ &\quad - \mu [R^{(2)} \cdot \bar{\mathbf{V}}_l + C^{(3)} : (\bar{\mathbf{V}}_l)(\bar{\mathbf{V}}_l)] \end{aligned} \quad [43]$$

The LHS of Eq. [43] can be expanded and put in a more familiar form by using Eq. [22] with the fact that motion of the interdendritic liquid attributed to the volume change is negligibly small (*i.e.*, neglect the area integral in Eq. [22]). The result is

$$\begin{aligned} \rho_l g_l \left[\frac{\partial \bar{\mathbf{V}}_l}{\partial t} + \bar{\mathbf{V}}_l \cdot \nabla \bar{\mathbf{V}}_l \right] &= -g_l \nabla P_l^l + \rho_l g_l \mathbf{g} + \nabla \cdot [\mu \nabla (g_l \bar{\mathbf{V}}_l) \\ &\quad + \mu \{\nabla (g_l \bar{\mathbf{V}}_l)\}^T \\ &\quad - (2\mu/3) \{\nabla \cdot (g_l \bar{\mathbf{V}}_l)\} I^{(2)}] \\ &\quad - \mu [R^{(2)} \cdot \bar{\mathbf{V}}_l + C^{(3)} : (\bar{\mathbf{V}}_l)(\bar{\mathbf{V}}_l)] \end{aligned} \quad [44]$$

VI. DARCY'S LAW AND PERMEABILITY

Darcy's law is valid under the following conditions:

(a) Slow flow; *i.e.*, $\bar{\mathbf{V}}_l \to 0$. Because of this, we can neglect the inertial term and second-order drag term in Eq. [44].
(b) Steady flow.
(c) The volume fraction of liquid, g_l, is uniform and constant.
(d) The forces of the liquid-liquid interactions are negligibly small.

With these conditions, Eq. [44] can be written in terms of $\mathbf{V}_l$, the superficial velocity (*i.e.*, the volume average velocity):

$$\mathbf{V}_l = -\frac{K^{(2)}}{\mu} \cdot [\nabla P_l^l - \rho_l \mathbf{g}] \quad [45]$$

where the permeability is defined as

$$K^{(2)} = g_l^2 [R^{(2)}]^{-1} \quad [46]$$

Equation [45] is Darcy's law.[32]

The more general form of the momentum equation (Eq. [44]) can also be written with the permeability and, if one chooses, with the average velocity of the interdendritic liquid, $\mathbf{V}_l^l$; then,

$$\rho_l g_l \left[\frac{\partial \mathbf{V}_l^l}{\partial t} + \mathbf{V}_l^l \cdot \nabla \mathbf{V}_l^l \right] = -g_l \nabla P_l^l + \rho_l g_l \mathbf{g} + \nabla \cdot [\mu \nabla (g_l \mathbf{V}_l^l) + \mu \{\nabla (g_l \mathbf{V}_l^l)\}^T - (2\mu/3) \{\nabla \cdot (g_l \mathbf{V}_l^l)\} I^{(2)}] - \mu \left[\frac{g_l^2}{K^{(2)}} \cdot \mathbf{V}_l^l + C^{(3)} : (\mathbf{V}_l^l)(\mathbf{V}_l^l) \right] \quad [47]$$

If the densities of the interdendritic liquid and the dendritic solid are equal and nonvarying, then the mass conservation equation (Eq. [25a]) reduces to

$$\nabla \cdot g_l \bar{\mathbf{V}}_l = 0 \quad [48]$$

Further, if the viscosity is nonvarying, then Eq. [37], combined with Eq. [48], reduces to

$$\nabla \cdot \left[(1/dV) \int_{dV} \sigma_l^{(2)} \gamma \, dv \right] = \mu \nabla^2 (g_l \bar{\mathbf{V}}_l) \quad [49]$$

or in terms of the superficial velocity, it is

$$\nabla \cdot \left[(1/dV) \int_{dV} \sigma_l^{(2)} \gamma \, dv \right] = \mu \nabla^2 \mathbf{V}_l \quad [50]$$

Finally, Eq. [47] simplifies to

$$\rho_l g_l \left[\frac{\partial \mathbf{V}_l^l}{\partial t} + \mathbf{V}_l^l \cdot \nabla \mathbf{V}_l^l \right] = -g_l \nabla P_l^l + \rho_l g_l \mathbf{g} + \mu \nabla^2 (g_l \mathbf{V}_l^l) - \mu \left[\frac{g_l^2}{K^{(2)}} \cdot \mathbf{V}_l^l + C^{(3)} : (\mathbf{V}_l^l)(\mathbf{V}_l^l) \right] \quad [51]$$

If it is further assumed that g_l is constant and uniform, then in terms of superficial velocity, Eq. [51] becomes

$$\rho_l \left[\left(\frac{1}{g_l} \right) \frac{\partial \mathbf{V}_l}{\partial t} + \left(\frac{1}{g_l^2} \right) \mathbf{V}_l \cdot \nabla \mathbf{V}_l \right] = -\nabla P_l^l + \rho_l \mathbf{g} + \left(\frac{\mu}{g_l} \right) \nabla^2 \mathbf{V}_l - \mu \left[\left(\frac{1}{K^{(2)}} \right) \cdot \mathbf{V}_l + \frac{C^{(3)}}{g_l^3} : (\mathbf{V}_l)(\mathbf{V}_l) \right] \quad [52]$$

Beckermann[20] derived an equation similar to Eq. [52], but he did not explicitly state that g_l is constant and uniform throughout the mushy zone. This is an important assumption in the context of alloy solidification. Even with this assumption, his equation differs from Eq. [52] in the term representing the liquid-liquid interaction, because he uses an effective viscosity in place of the viscosity of liquid. The origin of the effective viscosity, at least in the context of alloy solidification, is not evident.

VII. PERMEABILITY TENSOR AND SECOND-ORDER RESISTANCE COEFFICIENT

Let us assume that the permeability tensor in a dendritic single crystal (*i.e.*, an anisotropic medium) is symmetric so that it can be diagonalized.[26,35] That is, a set of three orthogonal directions (principal directions) exist for the medium along which the driving force ($\nabla P_l^l - \rho \mathbf{g}$) and velocity have the same direction. For the mushy zone during directional solidification of the single crystal, such preferred directions (principal directions) are along and perpendicular to the primary dendrites. In that case, the permeability tensor has three components (instead of six). A further simplification of the permeability tensor can be made for a columnar dendritic structure that is polycrystalline. Then there will be only two components of the permeability tensor, namely, a permeability parallel to the primary dendrites and a permeability normal to the primary dendrites. The models presented by Poirier[36] for permeabilities in columnar dendritic structures can be used along with the equations derived above. Finally, for equiaxial structures (fine-grained), we can assume that the permeability is isotropic.

Because the second-order resistance coefficient ($C^{(3)}$) is a third-order symmetric tensor, it can be shown[37] that for an isotropic medium, this coefficient vanishes (*i.e.*, equiaxial structures). For columnar dendritic structures, this coefficient does not vanish and has to be evaluated experimentally.

VIII. CONCLUSIONS AND RECOMMENDATIONS

1. In this analysis, the mushy zone is conceived as two interpenetrating phases; each phase is described with the usual field quantities, which are continuous in that phase but discontinuous over the entire space. On the microscopic scale, the usual conservation equation and the appropriate interfacial boundary conditions describe the state of the system. Because the solution of these equations in the microscopic scale is not practical due to the presence of highly complex interfacial geometries in the mushy zone, the scale at which the system is described is altered by averaging the microscopic equations over some representative elementary volume to yield macroscopic equations in terms of macroscopic field quantities. This volume-averaging technique is used to derive the mass and momentum conservation equations for the flow of interdendritic liquid during solidification.
2. The resistance offered by the solid to the flow of interdendritic liquid in the mushy zone is best described by two coefficients, namely, the reciprocal of permeability (*i.e.*, a first-order conductance coefficient) and a second-order resistance coefficient. Because equiaxial dendritic structures (fine-grained) can be assumed to be isotropic, the reciprocal of permeability alone is sufficient to quantify the resistance. On the other hand, for the flow in columnar structures, both coefficients are needed to quantify the resistance offered by the solid. For these structures, the second-order coefficient should be experimentally determined along with permeability.

3. To represent the conservation of mass and momentum for the flow of interdendritic liquid in the mushy zone with a stationary solid, Eqs. [25a] and [47] are recommended. To summarize, these two equations were derived with the following assumptions:
   (a) Densities of the dendritic solid and interdendritic liquid are uniform within the averaging region (dV).
   (b) Momentum exchange because of the volume changes during solidification is neglected.
   (c) Momentum dispersion is neglected.
   (d) Solid is stationary.
   (e) Interdendritic liquid is Newtonian.

   Further, if the densities of the interdendritic liquid and the dendritic solid are equal and nonvarying and the viscosity is nonvarying, then Eqs. [48] and [51] are recommended.
4. It can be seen that as $g_l \rightarrow 1$, then $\mathbf{V}_l^l \rightarrow \mathbf{v}_l$; hence, Eq. [25a] reduces to Eq. [20], which is the mass conservation equation for the single-phase liquid. In the case of the momentum equation, as $g_l \rightarrow 1$, the resistance coefficients vanish; *i.e.*, $K^{(2)} \rightarrow \infty$, and $C^{(3)} \rightarrow 0$. Again, Eq. [47] reduces to Eq. [27], which is the general momentum conservation for the single-phase liquid.

NOMENCLATURE

a^*	interfacial area between the liquid and solid within dV
$C^{(3)}$	third-order tensor resistance coefficient
d	characteristic length of dV (Figure 1)
dA	averaging area
dA_l	the part of dA occupied by liquid
da	microscopic elemental area
dV	averaging volume
dV_l	the part of dV occupied by liquid
dv	microscopic unit element
$\mathbf{F}$	total average force exerted by the liquid on the solid in dV
$\mathbf{g}$	gravitational acceleration
g_l	volume fraction of liquid
g_s	volume fraction of solid
$I^{(2)}$	second-order unit tensor
$K^{(2)}$	permeability (second-order tensor as defined by Eq. [46])
L	characteristic length of the mushy zone (Figure 1)
l	microscopic characteristic length (Figure 1)
$\mathbf{n}_{ls}$	unit normal to da^* pointing from the liquid toward the solid
$\mathbf{n}_{sl}$	unit normal to da^* pointing from the solid toward the liquid
P_l^l	average pressure (intrinsic volume average acting on the liquid in dV)
p_l	mechanical pressure on liquid element
$R^{(2)}$	second-order tensor resistance coefficient
$\mathbf{r}$	the position vector of a point in dV with respect to the fixed frame of reference (Figure 2)
S_l	areal fraction of the liquid
t	time
$\mathbf{V}_l$	superficial velocity (volume average)
$\mathbf{V}_l^l$	average interdendritic liquid velocity (intrinsic volume average)
$\bar{\mathbf{V}}_l$	mass-averaged velocity of interdendritic liquid
$\bar{\mathbf{V}}_s$	mass-averaged velocity of dendritic solid
$\mathbf{v}_l$	microscopic velocity of interdendritic liquid within dV
$\tilde{\mathbf{v}}_l$	deviation of $\mathbf{v}_l$ from $\bar{\mathbf{V}}_l$
$\mathbf{v}_s$	microscopic velocity of dendritic solid within dV
$\mathbf{w}$	velocity of the microscopic liquid-solid interface
$\mathbf{x}$	the portion vector of the centroid of dV with respect to the fixed frame of reference (Figure 2)
Γ	tortuosity function defined by Eq. [40]
γ	liquid volume distribution function
$\boldsymbol{\xi}$	the position vector of a point in dV with respect to the centroid of dV (Figure 2)
μ	viscosity of the interdendritic liquid
ρ_l	mass density of liquid
ρ_l^l	intrinsic volume-averaged mass density of the liquid
$\sigma_l^{(2)}$	deviatoric part of the general material stress tensor of the liquid element
$\bar{\Phi}_l$	mass average of ϕ_l
ϕ_l	any microscopic quantity of the liquid
$\tilde{\phi}_l$	the deviation of ϕ_l from its mass average value
Ψ_l	volume average of ψ_l
Ψ_l^l	intrinsic volume average of ψ_l
$\bar{\Psi}_l$	mass average of ψ_l
ψ	any microscopic quantity
ψ_l	any microscopic quantity of the liquid
ψ_s	any microscopic quantity of the solid
$\tilde{\psi}_l$	the deviation of a microscopic quantity from its mass-averaged value of the liquid
$\bar{\bar{\psi}}_l$	mass average of $\tilde{\psi}_l$

ACKNOWLEDGMENTS

The authors are grateful for the partial support of this work through a grant from Alcoa Technical Center and Grant No. NAG 3-723 from the Microgravity Science and Applications Division of NASA. Professor C.L. Chan of the Department of Aerospace and Mechanical Engineering, University of Arizona, Tucson, deserves special thanks, because he shared his knowledge of pertinent literature with us when we initiated this work. The authors have had numerous discussions pertaining to the representation of the momentum equation for dendritic solidification with Dr. P. Nandapurkar and Professors J.C. Heinrich and A.J. Pearlstein of the University of Arizona and with Dr. W. Bennon of Alcoa Technical Center, Alcoa Center, PA. We sincerely thank each of them for sharing their insights with us.

REFERENCES

1. M.C. Flemings: *Solidification Processing*, McGraw-Hill, New York, NY, 1974, pp. 244-46 and 83-85.

2. M.C. Flemings and G.E. Nereo: *Trans. TMS-AIME*, 1967, vol. 239, pp. 1449-61.
3. M.C. Flemings, R. Mehrabian, and G.E. Nereo: *Trans. TMS-AIME*, 1968, vol. 242, pp. 41-49.
4. M.C. Flemings and G.E. Nereo: *Trans. TMS-AIME*, 1968, vol. 242, pp. 50-55.
5. R. Mehrabian, M. Keane, and M.C. Flemings: *Metall. Trans.*, 1970, vol. 1, pp. 1209-20.
6. S. Kou, D.R. Poirier, and M.C. Flemings: *Electric Furnace Proc.*, ISS-AIME, Chicago, IL, 1977, vol. 35, pp. 221-28.
7. T. Fujii, D.R. Poirier, and M.C. Flemings: *Metall. Trans. B*, 1979, vol. 10B, pp. 331-39.
8. S.D. Ridder, S. Kou, and R. Mehrabian: *Metall. Trans. B*, 1981, vol. 12B, pp. 435-47.
9. J. Szekely and A.S. Jassal: *Metall. Trans. B*, 1978, vol. 9B, pp. 389-98.
10. W.D. Bennon and F.P. Incropera: *Int. J. Heat Mass Transfer*, 1987, vol. 30, pp. 2161-70.
11. M. Hassanizadeh and W.G. Gray: *Adv. Water Resour.*, 1979, vol. 2, pp. 131-41.
12. W.G. Gray: *Chem. Eng. Sci.*, 1975, vol. 30, pp. 229-33.
13. J.C. Slattery: *AIChE J.*, 1967, vol. 13, pp. 1066-71.
14. S. Whitaker: *Chem. Eng. Sci.*, 1973, vol. 28, pp. 139-47.
15. T.B. Anderson and R. Jackson: *Ind. Eng. Chem. Fundam.*, 1967, vol. 6, pp. 527-39.
16. D.A. Drew and L.A. Segel: *Stud. Appl. Math.*, 1971, vol. 50, pp. 233-57.
17. G. Ahmadi and M. Farshad: *Indian J. Technol.*, 1974, vol. 12, pp. 195-98.
18. J.A. Trapp: *Int. J. Eng. Sci.*, 1976, vol. 14, pp. 991-98.
19. S. Whitaker: *Adv. Heat Transfer*, 1977, vol. 13, pp. 119-203.
20. C. Beckermann: Ph.D. Thesis, Purdue University, West Lafayette, IN, 1987.
21. A. Bejan: *Handbook of Single-phase Convective Heat Transfer*, S. Kakae, R.K. Shah, and W. Aung, eds., John Wiley & Sons, New York, NY, 1987, pp. 16.1-16.34.
22. J.G. Georgiadis and I. Catton: *Trans. ASME*, 1986, vol. 108, pp. 284-90.
23. F. Chen and C.F. Chen: *J. Heat Transfer*, 1988, vol. 110, pp. 403-09.
24. V.R. Voller and C. Prakash: *Int. J. Heat Mass Transfer*, 1987, vol. 30, pp. 1709-19.
25. C. Beckermann and R. Viskanta: *Physico Chem. Hydrodyn.*, 1988, vol. 10, pp. 195-213.
26. S. Whitaker: *Ind. Eng. Chem.*, 1969, vol. 61, pp. 14-28.
27. W.G. Gray and P.C.Y. Lee: *Int. J. Multiphase Flow*, 1977, vol. 3, pp. 333-40.
28. J.C. Slattery: *AIChE J.*, 1969, vol. 15, pp. 866-72.
29. S. Whitaker: *AIChE J.*, 1967, vol. 13, pp. 420-27.
30. G.K. Batchelor: *An Introduction to Fluid Dynamics*, Cambridge University Press, Cambridge, 1967, pp. 141-42.
31. R.J. McDonald and J.D. Hunt: *Trans. TMS-AIME*, 1969, vol. 245, pp. 1993-97.
32. R.B. Bird, W.E. Stewart, and E.N. Lightfoot: *Transport Phenomena*, John Wiley, New York, NY, 1960, pp. 59 and 150.
33. G.K. Batchelor: *An Introduction to Fluid Dynamics*, Cambridge University Press, Cambridge, MA, 1967, p. 147.
34. W.G. Gray and K. O'Neill: *Water Resour. Res.*, 1976, vol. 12, pp. 148-54.
35. A.E. Scheidegger: *The Physics of Flow Through Porous Media*, Macmillan Company, New York, NY, 1960, pp. 76-79.
36. D.R. Poirier: *Metall. Trans. B*, 1987, vol. 18B, pp. 247-55.
37. R. Aris: *Vectors, Tensors and the Basic Equations of Fluid Mechanics*, Prentice Hall, Englewood Cliffs, NJ, 1962, p. 31.